

Problem definition

Problem. Hateful content online increasingly travels as video, where meaning is carried jointly by visual context, audio delivery, and the spoken message. Single-modality classifiers struggle on cases where these three signals must be combined to distinguish hate from satire or critical discourse.

Hypothesis. A small fusion transformer trained on top of three frozen encoders (vision, audio, text) can outperform every single-modality baseline, but the gains are not uniform across metrics or communities.

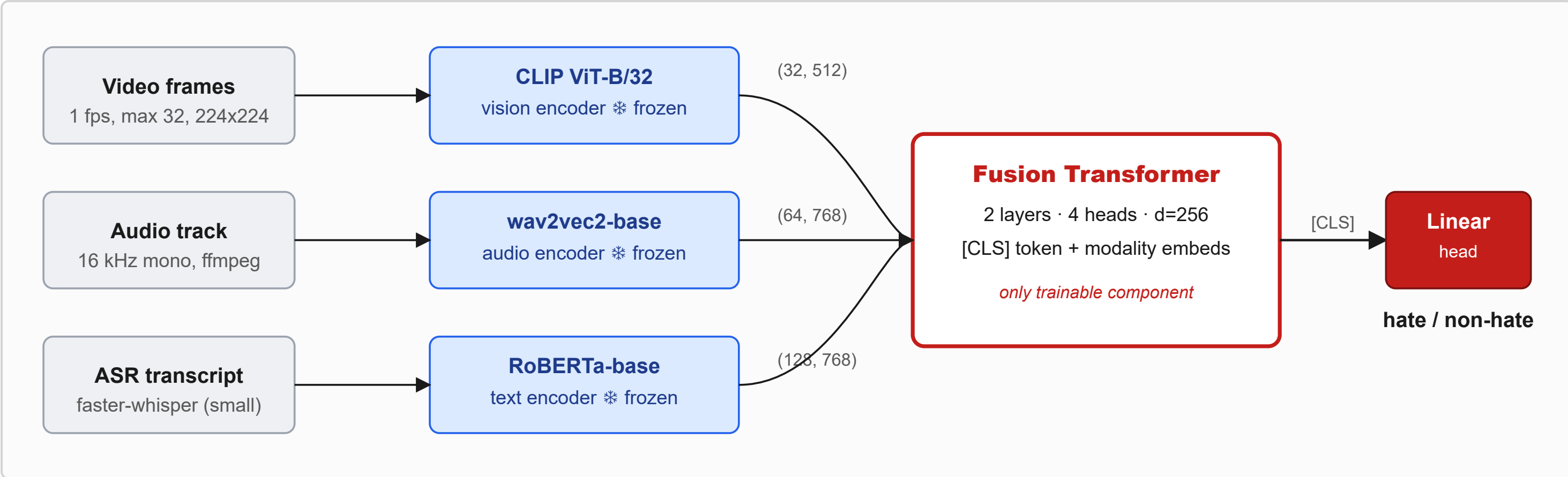
Dataset

- **HateMM [1]: 1,083 BitChute videos** (431 hate, 652 non-hate); ~43.5 hours, 6.3 GB. CC BY 4.0.
- A video is labeled *hate* when it attacks or demeans someone for who they are: religion, race, ethnicity, gender, sexual orientation, or disability. Each hate video also tags the targeted community.
- 80/10/10 stratified split by binary label (863/109/109), fixed seed; preprocessing succeeded on 1081/1083 videos.
- **Scope:** BitChute is a low-moderation platform, so our results characterise hate detection on that source distribution; transfer to YouTube or TikTok would require recalibration.

Method

1. **Preprocess.** Sample frames at 1 fps capped at 32 per video, center-cropped to 224 x 224; extract 16 kHz mono audio via ffmpeg; transcribe speech with faster-whisper.
2. **Frozen feature extraction.** CLIP ViT-B/32 per frame, wav2vec2-base mean-pooled into 64 audio tokens, RoBERTa-base token embeddings (length 128) on the transcript.
3. **Fusion.** All token sequences are concatenated, prepended with a [CLS] token and per-modality type embeddings, then fed into a 2-layer pre-norm transformer (d = 256, 4 heads, GELU). The [CLS] output goes to a binary linear classifier.

Why frozen encoders. Keeping all three encoders frozen makes the ablation fair: only the fusion head changes across configurations. Running the same head with one modality at a time gives each unimodal baseline.



Related works

- **HateMM [1].** Defines the task and releases the dataset and target-community labels we audit against; their best multimodal baseline reaches ~0.80 macro-F1.
- **Hateful Memes [2].** Foundational multimodal hate benchmark (image + text); MOMENTA [3] adds context-aware fusion on the same line of work.
- **Attention Bottlenecks (Nagrani et al.) [4].** Show that cross-modal attention can match heavy late-fusion pipelines at lower cost; we use a lighter variant suited to our dataset size.
- **Large pretrained encoders [5, 6, 7]** transfer well to downstream tasks with only a small head trained on top; the frozen-encoder paradigm we rely on here.

The Seed Trap

Our first run (seed 42 only) showed fusion beating text by +5.5 accuracy points , a headline result. Re-running across three seeds {42, 1337, 2025} made the gap vanish completely .				
Model	Seed 42 alone		3-seed mean ± std	
Fusion (V+A+T)	83.5% acc	F1 0.824	79.2% ± 5.2%	F1 0.782 ± 0.055
Text only	78.0% acc	F1 0.763	79.2% ± 2.1%	F1 0.781 ± 0.021
Gap	+5.5 pp		0.0 pp, dead tie	
Only AUROC survived multi-seed scrutiny (0.876 vs 0.827). Single-seed reporting would have published a false claim.				

Validation

Table 1. Modality ablation on the test split (n = 109), mean ± std over 3 seeds {42, 1337, 2025}. Our 0.78 macro-F1 is comparable to Das et al.'s reported ~0.80 on the same dataset.

Modality	Acc	F1	Prec	Rec	AUROC
V + A + T	0.79 ± 0.05	0.78 ± 0.05	0.74 ± 0.08	0.74 ± 0.08	0.88 ± 0.01
V + A	0.77 ± 0.01	0.76 ± 0.00	0.71 ± 0.04	0.71 ± 0.08	0.85 ± 0.01
V + T	0.78 ± 0.03	0.77 ± 0.02	0.73 ± 0.07	0.72 ± 0.04	0.86 ± 0.01
A + T	0.76 ± 0.02	0.75 ± 0.02	0.71 ± 0.05	0.69 ± 0.08	0.84 ± 0.00
Text only	0.79 ± 0.02	0.78 ± 0.02	0.75 ± 0.06	0.73 ± 0.08	0.83 ± 0.02
Video only	0.72 ± 0.03	0.71 ± 0.02	0.64 ± 0.06	0.68 ± 0.06	0.81 ± 0.01
Audio only	0.67 ± 0.02	0.67 ± 0.03	0.56 ± 0.02	0.71 ± 0.05	0.77 ± 0.03

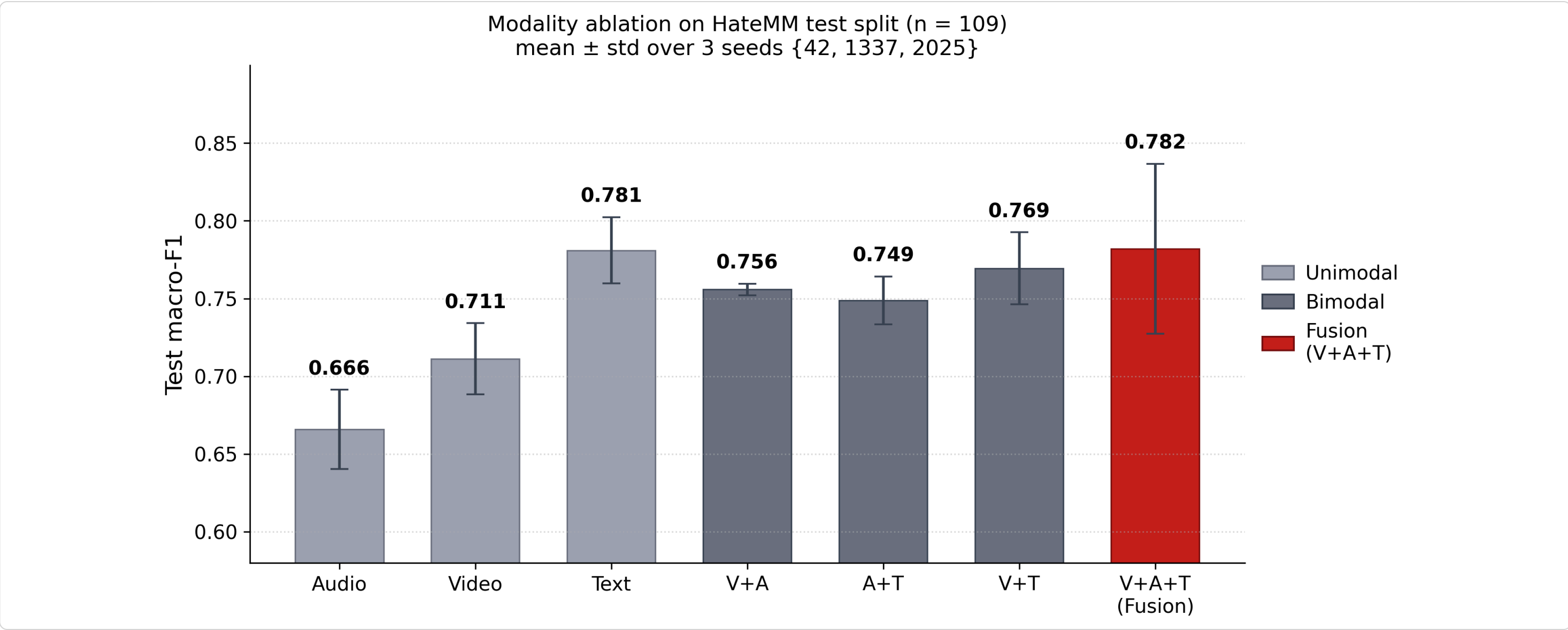


Figure 1. Test macro-F1 for all seven modality configurations, mean ± std over 3 seeds {42, 1337, 2025}.

Key observations

Each added modality lifts AUROC monotonically. From audio alone (0.77) to full fusion (0.88), every step in the seven-row ablation buys ranking signal. Accuracy, in contrast, plateaus at the text-only level (0.79).

Video is the stronger complement to text. Text → V+T adds +0.030 AUROC; text → A+T adds only +0.012. Adding the third modality on top of any pair gains another ~+0.02.

Fairness Analysis

Table 2. Per-community test accuracy (n ≥ 15), mean ± std over 3 seeds.

Community	n	Fusion	Video	Audio	Text
Blacks	46	0.75 ± 0.05	0.70 ± 0.04	0.59 ± 0.06	0.73 ± 0.03
Others	39	0.84 ± 0.01	0.74 ± 0.09	0.68 ± 0.11	0.85 ± 0.05
Jews	15	0.73 ± 0.13	0.67 ± 0.07	0.71 ± 0.04	0.76 ± 0.04

⚠ **Others (n = 39)** is predominantly non-hate in our test split: high accuracy reflects majority-class prediction, not genuine discrimination. Text wins on *Others* and *Jews*; fusion leads only on *Blacks*. No single model is best across all communities.

Limitations

- **Small test set (n = 109).** Per-community stats for groups with n < 15 are excluded; reported groups still carry appreciable variance.
- **Platform specificity.** Models trained on BitChute may not generalise to lower-toxicity platforms where the visual and linguistic style of hateful content differs substantially.
- **ASR quality varies.** Shouting, music, and accented speech degrade the text modality on a subset.
- **Frozen backbones cap performance.** Audio is compressed to 64 fixed slots by wav2vec2 mean-pooling, losing detail for long videos. Fine-tuning even one encoder would improve absolute numbers but break the controlled ablation.

Conclusion

Fusing three frozen encoders through a small transformer head **robustly improves AUROC** over every unimodal baseline (0.88 ± 0.01 vs 0.83 ± 0.02 for text alone). On accuracy and F1 the gain disappears once you run three seeds, something single-seed reporting would have missed entirely. Fusion improves how well the model *ranks* videos by risk; it does not improve how many it classifies correctly.

Practical implication. Three deployment options on this dataset, choose by your cost/quality trade-off: (i) **V+A+T (AUROC 0.88 ± 0.01)** when ranking quality matters most; (ii) **V+T (AUROC 0.86 ± 0.01)** as a cost-aware sweet spot, within 0.02 AUROC of full fusion with no audio pipeline at all, capturing ~60% of the AUROC lift over text-only; (iii) **Text-only (acc 0.79 ± 0.02)** for fixed-threshold binary decisions, the cheapest pipeline that still ties V+A+T on accuracy.

References

1. M. Das et al., "HateMM," *ICWSM*, 2023.
2. D. Kiela et al., "Hateful Memes Challenge," *NeurIPS*, 2020.
3. S. Pramanick et al., "MOMENTA," *EMNLP Findings*, 2021.
4. A. Nagrani et al., "Attention Bottlenecks for Multimodal Fusion," *NeurIPS*, 2021.
5. A. Radford et al., "CLIP," *ICML*, 2021.

6. A. Baevski et al., "wav2vec 2.0," *NeurIPS*, 2020.
7. Y. Liu et al., "RoBERTa," *arXiv:1907.11692*, 2019.



Scan for code
github.com/YDIB11/Mini-Project
EE-559 Deep Learning · EPFL 2026